DEVELOPMENTS AND LIMITATIONS IN MAPPING PHYSIOLOGICAL REACTIONS TO CHANGES IN MUSICAL STRUCTURE

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ABSTRACT

Musical affect experiments have shown that common physiological markers, such as heart rate, brain waves, skin conductance, and respiratory variance, are all affected by varying musical forms and structures (Ellis and Brighouse 1952).

Using a battery of biofeedback sensors, a neural network model can be trained to map physiological changes to the structural variables of a music generation algorithm. Once trained, the model can be used to control the music algorithm, predicting the response of the listener to the music generated.

This paper discusses relevant developments and limitations, including the expanding range of available biofeedback technologies, electroencephalogram (EEG), approaches to controlling external variables, consideration of individual responses and cultural backgrounds, social and commercial applications of automated music generation, creative ramifications for composers and performers, and methods for measuring success in such an undertaking.

1. INTRODUCTION

By exposing participants to a musical algorithm that uses randomised musical structures, and comparing the musical structures to the measured physiological changes in the participants, we are developing a system which can generate new musical material with enough foresight to invoke any requested human physiological change.

In order to build such a system, several different technologies must be successfully brought together:

- EEG and biosensor technologies
- A generative music algorithm with a broad range of music structures
- Data collection using participants
- A neural network
- A definition for success

1.1. Limitations

Recorded physiological reactions may be related to any underlying mental or bodily mechanic, but by controlling the external environment sufficiently, variations from a reasonably stable and well established neutral state can be obtained. Thus, if neutral states are observed appropriately, results are more likely to be a result of the biological musical phenomenon of the individual, (D. Ellis and Brighouse 1952). The project also relies on minimizing the irrelevant physical and mental factors affecting the participants, and managing conditions for the participants so as to promote engagement with the music.

Each model is trained to use a particular music algorithm. These music algorithms are simple, but are required to encompass a broad number of musical structures. Too many variables can create an almost limitless number of possible combinations, thus minimizing variables in musical algorithms is necessary.

1.2. Scope

From the point of view of a sonic arts researcher, whose specialisation is in electronic music composition, there are also inherent limitations of scope, such as expertise limitations in the multiple fields involved, ethical considerations, and technical limitations. It should be observed that the aim of this project is to use established and well documented technologies to create a composition system, not to pioneer new technologies.

2. BIOSENSORS

The value of different biosensors was assessed for musical relevance according to literature, practicality, non-invasiveness, ethics, and required expertise, in order to shortlist several appropriate biosensors. Developments in amateur electronic and consumer products continue to make such technologies increasingly accessible to those outside of specialist medical fields.

2.1. Electroencephalogram



Figure 1. Wearing the Emotiv Epoc

For electroencephalography (EEG), the *Emotiv Epoc*, with 14 electrodes, simple implementation, and massively powerful software suite was chosen over tedious and expensive medical grade counterparts, (Lang 2012; Duvinage et al. 2013; Badcock et al. 2013; "Legitimising Affective Suite" 2014; Miranda 2010; Miranda et al. 2003). Using EEG data alone has already shown some predictive potential. Collura provides an excellent overview of EEG technology, (Collura 1993).

2.2. Arduino Biosensors



Figure 2. Wearing the biosensors

Biosensors implemented using an Arduino are being investigated:

Heart rate and variance is measured using a simple open source Arduino pulse monitor, (See "PulseSensor - Pulse Sensor Amped!" 2014). Heart rate variance has been shown to vary greatly in relation to tempo and rhythmic variations, (Bernardi, Porta, and Sleight 2006; D. Ellis and Brighouse 1952; R. J. Ellis et al. 2009)

Galvanic skin response, famous for its use in lie detectors, is also implemented with two surface electrodes on the fingers, and a simple circuit, (Woolley-Hart 1972; Lykken and Venables 1971; Coutinho and Cangelosi 2011).

Respiratory rate has been shown to vary with musical stimulus, (Siwiak, Berger, and Yang 2009; Bernardi, Porta, and Sleight 2006; D. Ellis and Brighouse 1952). Respiratory rate and size is measured using belt with conductive rubber cord around the chest, which increases in resistance when stretched, (Coyle et al. 2009).

Finally, body temperature is easily measured using a single temperature sensitive component, the *TMP36,* (See "TMP36 Datasheet and Product Info" 2014).

There are a many other suitable biosensors that can be investigated for such a purpose, such as pupillary response, (Schlemmer et al. 2005), or even facial expression recognition, (Bettadapura 2012).

3. A GENERATIVE MUSICAL ALGORITHM

The musical algorithms are designed to produce music with varying (but controlled) musical structures, including time signatures, scales, set rhythms, forms, and where applicable, timbre, (Patil et al. 2012).

The algorithms, written in supercollider, accept an array of numbers, each index being a variable for a specific structural aspect of the music generated. The focus is achieving a broad number of structural possibilities, with as few variables as possible. Simple mathematical statements reliably recreate each stream of music, rather than long markov chains, chance, predefined blocks of notes, or any kind of self-analysis, which could involve hundred if not thousands of variables.

The algorithms are built to require as few variables as possible to minimize the large number of possible outcomes, yet aim to maximise the clarity of the musical structure. Initially, without a trained neural network model selecting pieces, the majority of the musical results are un-interesting, and often not musical at all.

4. DATA COLLECTION WITH PARTICIPANTS

Healthy normal participants are selected randomly, from both genders, both musically trained and untrained.

Each participant is given instructions to avoid participating on days of extreme stress, tension, or under the influence of any significant drugs or sleeplessness. Visual and aural distractions and interruptions are minimised, and air conditioning is set to a neutral temperature.

Once the biosensors are fitted to the participant, some time listening to music is spent without recording response data, in order to minimise the influence of situational novelty. Before and after each passage, a silent period of 20-30 seconds of bio-feedback is recorded in order to establish a neutral level. A single music passage is played (with randomised structure) in repeats for between 45 seconds and 2 minutes depending on its length and tempo.

To keep participants engaged in listening, participants are monitored for signs of boredom or sleepiness, and take frequent breaks. Generally a participant can't fully engage with listening for more than 15 minutes without a break.

The biosensor data is processed with Cycling 74's Max. The EEG software also analyses the EEG data, defining the proprietary 'affective' measurements of engagement, meditation, frustration, and excitement, (See "Legitimising Affective Suite" 2014). The results of both these analyses are entered alongside the musical array that stimulated it into an SQL database. The raw data from each sensor (and each EEG electrode) is also kept in a key-named file to allow for deeper analysis later.

5. A NEURAL NETWORK

Feed forward artificial neural networks, well established as universal approximators, are uniquely suited to analysing this particular problem, (Hornik, Stinchcombe, and White 1989). Neural networks excel at finding complex relationships in data with a lot of variables, such as diagnosing potential cancer patients, (Bottaci et al. 1997), or predicting rainfall, (Wu and Chen 2009).

5.1. Architecture

The example shown in figure 3 and 4 involves data from an EEG-only experiment. A demonstration of the music produced from this experiment can be heard on Sound Cloud, (See Pitman 2014).

A neural network's nodes are divided into input layers, hidden layers, and output layers. Each input coincides with structural variable from the database, and the output nodes are associated with each physiological

variable. In this example the 5 'affective' values given by the EEG software's inbuilt analysis are used as the outputs.

Neural Network Model Summary (SPSS example)	
Structure	
129 Inputs	
1st Hidden Layer 8 nodes	
2nd hidden layer 5 nodes	
Output layer 5 nodes.	
Sample Structure	
348 Total samples	
80% Training	
10% Testing	
10% Hold Out	

Error Summary	
Holdout Average Overall Relative Error	.279
Relative Error for Dependents	
frust	.340
medit	.177
exctShort	.273
exctLong	.368
engagement	.296

Figure 3. Neural Network Architecture

5.2. Training

As matching structures and reactions are entered repeatedly into the inputs and outputs of the network, successful mathematical pathways through the hidden layers are reinforced with each sample. With more samples, trends become more prominent. With each repeat, the model changes less and less until a relatively stable state is established, at which point the training is complete, and the model can make predictions.

6. MEASURING SUCCESS

The trained neural network model can now make predictions about the changes to the affective variables. This gives us the ability to search for appropriate musical passages without endlessly rendering and auditioning results.

We can derive a notion of success by simply listening to the results, however this only provides a rough and potentially biased measure, nor does it take into account the individuals dynamic mental state, environment, or time since the samples were recorded.

A more reliable measure of success can be established by withholding some of the data samples, and comparing the model's predictions to the known results. As these withheld results are from the same time period, environment, and individuals, they provide a relevant comparison.

In figure 3, the relative error for each affective value is given. The graph in figure 4 compares predictions on the *y* axis, to the known results on the *x* axis for each affective value. Variation from the line of $x=y$ implies error. While some variation is seen, a general trend is also apparent, indicating some success.

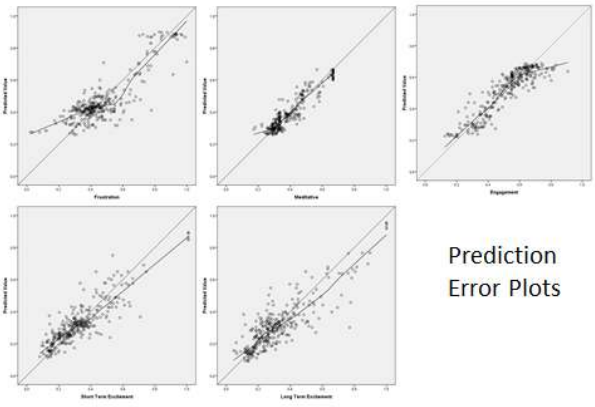


Figure 4. Prediction Error Plots

6.1. A Context for 'Success'

It is straightforward to suggest that there is no one correct answer to asking, "How will people react to this music?" Thus, we could argue that some prediction error, which might hinder a laboratory scientist, might be considered 'appropriate' here.

We can consider these results as an extrapolation of the participants as individual entities, or as a unique group, at a particular time in their lives. Music generated in this way inherits a human context, despite being generated algorithmically.

6.2. Composition

The composer now has the ability to compose music in a novel medium, where only the physiological end result needs to be specified, and music is generated accordingly. This has some interesting performance and composition ramifications, especially when combined with performers wearing biosensors, driving the system in real-time. A score might be as simple as having a performer wear biosensors,

- Think of the worst moment of your life for 3 mins,
- Breathe very fast for 30 seconds,
- Look at the provided images of puppies for 30 seconds,
- Think of a very embarrassing fact about your childhood for 20 seconds,
- Tell the audience your embarrassing fact,
- Hold a water pistol to your temple for 10 seconds, pull the trigger…

While the algorithm and neural network are tasked with generating music in real-time to match or counter the physiological states of the performer.

6.3. Implementation

A system such as this could be designed to generate appropriate music in real-time for games, movies, or other media, following instructions passed from the media itself, (e.g. high frustration, sweaty palms)

This system could also be implemented for medical or therapeutic uses, even surgery, if for example, lowering heart rate is absolutely essential to an operation.

While certain effects are well documented and succinct, such as the relationship between tempo and heart rate variance, other effects are far more elusive and complex, (Hevner 1937). This type of system might prove an invaluable resource for understanding the biological musical phenomenon.

7. FUTURE DEVELOPMENTS

While success in producing art or music can be argued, striving to reduce predictive error and is still essential to the concept. Further development in this area is required before the system could be implemented practically.

The aim is to not only decrease the prediction error to within more statistically acceptable margins, but also to allow more complicated and engaging executions of generated music.

It is likely that future generations of this system will be implemented for use as programming libraries or with APIs that will allow the system to be imbedded inside media projects such as games or movies.

As development continues, more biosensors, better auditioning procedures, other methods of analysis and prediction, and refinements to the music algorithm are investigated.

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